

A hard-working team player with proven problem-solving ability

Technical Skills

Programming Languages	Experienced with: C, C++, MATLAB, and Python3 Familiar with: Bash, C#, Java, Javascript, and PHP
Development Tools	Experienced with: Git, Docker, Jira, and UML Familiar with: CMake, Jenkins, JUnit, Make, Nodejs, TeamCity, and Tensorflow
Operating Systems	Experienced with: macOS, Ubuntu Linux, and Windows 7+

Education

Jul 2019 Dec 2022 (anticipated)	Doctor of Philosophy – Engineering , <i>The University of Melbourne, Australia</i> • Field: Electrical and Electronic Engineering • Thesis title: Adversarial Learning Games for IoT and Cognitive Radios • Coursework: Optimisation and machine learning
Feb 2015 Dec 2018	Bachelor of Engineering , <i>University of Canterbury, New Zealand</i> • Field: Computer Engineering (GPA: 8.66 / 9.00) • Minor: Communications and Network Engineering • Coursework: Software Engineering fundamentals, algorithms, OSs, and networks

Publications

2021	First author , <i>Generative Adversarial Networks for anomaly detection on decentralised data</i> in Annual Reviews in Control, doi: 10.1016/j.arcontrol.2021.10.002 • An expanded version of our 2020 work with GANs
2021	First author , <i>Privacy-Preserving Collaborative SDR Networks for Anomaly Detection</i> in IEEE International Conference on Communications (ICC), doi: 10.1109/ICC42927.2021.9500807 • An empirical study of parameter-sharing techniques to collaboratively train autoencoders
2020	First author , <i>Distributed Generative Adversarial Networks for Anomaly Detection</i> , in Decision and Game Theory for Security, doi: 10.1007/978-3-030-64793-3_1 • A competitive anomaly detector in collaborative settings using GANs for collaborative training
2020	Fourth author , <i>Designing Neural Networks for Real-Time Systems</i> , in IEEE Embedded Systems Letters, doi: 10.1109/LES.2020.3009910 • A performance analysis of a neural network converter for which I was the main programmer

Work Experience

Feb 2022 Dec 2022 (anticipated)	Software Developer , <i>Orbital Transports Inc, USA</i> • Writing an iOS app and Laravel webapp to interface with a satellite in low Earth orbit that: ◦ Will be designed based on my submission to the Idea-to-Space Launch Contest ◦ Has an expected launch date of Oct 2022
Mar 2021 Present	Subject Demonstrator , <i>The University of Melbourne, Australia</i> • Marking exams and overseeing workshops for System Optimisation and Machine Learning: ◦ Helping develop workshop content in Python3 using Jupyter, pyomo and Tensorflow ◦ Assisting 50+ students with optimisation and programming problems
Apr 2020 Feb 2022	(Volunteer) Software Engineer , <i>Melbourne Space Program, Australia</i> • Worked with the AI and Robotics team dedicated to the University Rover Challenge • Wrote a ROS-based autonomous navigation system with routing and obstacle detection
Feb 2020 Dec 2020	Tutor , <i>Vz s College, Australia</i> • Provided one-on-one help to students in the following subjects: <i>Mathematics, Physics, Chemistry, Biology, and English</i>

Nov 2018
Feb 2019

Research Intern, *The University of Auckland, New Zealand*

- Wrote a converter tool to turn Keras neural networks into pipelined, multicore, time-predictable C code. The tool was:
 - Written in Python3 using Jinja2 templating
 - Converted by teammates into an online tool, aitransformer.com

Nov 2017